Tut3

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Part I
The discriminant

Let K be a number field with embeddings

$$\sigma_1, .., \sigma_n : K \to \mathbb{C}$$

For $\alpha_1, .., \alpha_n \in K$ the **discriminant** is

$$disc(\alpha_1, .., \alpha_n) = |\sigma_i(\alpha_j)|^2$$

Put another way, if $(\sigma_1, ..., \sigma_n) : K \to \mathbb{C}^n$, we get the

0.0.1 prop

The discriminant is

$$|T(\alpha_i \alpha_j)|$$

Proof: $T = \sum \sigma_k(-)$, and $det([\sigma_i(\alpha_j)]^2)$is exactly that.

0.0.2 corollary

$disc(\alpha_1, .., \alpha_n) \in \mathbb{Q}$, and if all α_i are algebraic integers, then $disc(\alpha_1, .., \alpha_n) \in \mathbb{Z}$.

0.0.3 theorem

$disc = 0$ iff $\alpha_1, .., \alpha_n$ are linearly dependent. Indeed if $\alpha_1, .., \alpha_n$ are dependent, then the determinant will vanish. Conversely, if $\alpha_1, .., \alpha_n$ span $K/\mathbb{Q}$, since the embeddings $K \to \mathbb{C}$ are linearly independent by Artin's independence of characters, the determinant can't vanish.

Another proof, using the $|T(\alpha_i \alpha_j)|$ perspective, is that if the rows of $[T(\alpha_i \alpha_j)]$ were dependent over $\mathbb{Q}$ with coefficients $a_1, .., a_n$, letting

$$\alpha = \sum a_i \alpha_i$$

we'd get $T(\alpha \alpha_j) = 0$ for all j, but $\alpha \alpha_j$ span K over $\mathbb{Q}$, so the trace would be degenerate.

0.0.4 Formula for polynomial ring

If $K = \mathbb{Q}[\alpha]$ and $\alpha_1, .., \alpha_n$ are the **conjugates** of $\alpha \in \mathbb{Q}$ (where one of them is α) then

$$disc(1, \alpha, .., \alpha^{n-1}) = \prod_{1 \leq r < s \leq n} (\alpha_r - \alpha_s)^2 = \pm N^K(f'(\alpha))$$

for f the monic irreducible polynomial for α over $\mathbb{Q}$, the $+$ corresponds to $n = 0, 1 \, mod \, 4$.

Proof:

The matrix $[\sigma_i(\alpha^{j-1})| = |\alpha_i^{j-1}|$ is Vandermonde on $(\alpha_1, .., \alpha_n)$, which has determinant

$$\prod_{r<s} (\alpha_r - \alpha_s)$$

For the derivative formula,

$$f(x) = \prod (x - \alpha_i)$$

so

$$f' = \sum_i \frac{f}{(x - \alpha_i)}$$

and

$$f'(\alpha) = \prod_{\alpha_i \neq \alpha} (\alpha - \alpha_i)$$

taking the norm, we get

$$\prod_i \prod_\sigma (\sigma(\alpha) - \sigma(\alpha_i))$$

as desired.

Part II
The Vandermonde formula

An equivalent approach to the one Ian wrote out is that the determinant of $[z_i^j]$ (thought of as a polynomial in the variables z_i) must be divisible by $(z_i - z_j)$, which has the correct degree $\binom{n}{2} = \sum_1^{n-1} i$, and so is the determinant up to a constant factor. He then explained how to compute the constant factor inductively.

Part III
Representations of R

In modern math, we try to reduce any problem to "linear algebra", or as make as linear as we can.

Some examples are:

1. In manifold we study M via the **tangent bundle** $\mathcal{T}_M$**,** or generally **sheaves** on M.
2. In group theory, we study G via the **representations** of G.
3. In ring theory, we study R via $Mod(R)$, the modules on which R act.

In all of these, often the association $X \mapsto Linearization(X)$ isn't injective, but it's a fruitful first approximation.

Some examples of modules $M \in R$ are given via ideals $I \subset R$.

0.0.5 Definition

We say two ideals I_1, I_2 are **equivalent** if $I_1 \cong I_2$ as abstract $Mod(R)$ modules.

0.0.6 Examples

- For $R = \mathbb{Z}$, then $(p\mathbb{Z}) \cong (q\mathbb{Z}) \cong \mathbb{Z}$ for every prime p, via
$$p\mathbb{Z} \to \mathbb{Z}$$
$$x \mapsto x/p$$
- For $R = k[x]$, then $(x) \cong k[x]$ via
$$(x) \to k[x]$$
$$y \mapsto y/x$$

In a slogan: $(p\mathbb{Z})$ doesn't know intrinsically that its elements are divisible by p.

We will see examples of R and ideals I which are **not** isomorphic to R abstractly.

0.0.7 Prop

Let R be an integral domain, $I, J \subset R$ ideals (in our case, $R = \mathcal{O}_K$ a ring of integers), then if there exists $\alpha \in Frac(R)$ with
$$\alpha I = J$$
then $I \cong J$ as abstract R modules.

Proof: We define a map
$$I \to J$$
$$i \mapsto \alpha i$$
that by assumption is an equivalence.

0.0.8 Theorem

For R an integral domain, $I \cong J$ if and only if there exists such an α.

Proof: If $f : I \to^{\cong} J$ then $f(ii') = if(i') = i'f(i)$ and so $f(i)/i = f(i')/i'$ as desired.

0.0.9 Definition

The equivalence classes of ideals are called "ideal classes".

0.0.10 Definition

Given two ideals I, J in an integral domain R, their product is $IJ :=< ij | i \in I, j \in J >$.

0.0.11 Prop

The product descends to a well defined product on equivalence of ideals.

Proof: If $\alpha I \cong I'$, then $\alpha IJ = I'J$, and so $IJ \cong I'J$.

0.0.12 Theorem

For R a ring of integers, as abstract R modules,

$$IJ = I \otimes_R J$$

Proof: We might discuss this in the future.

Note that in this perspective, the product is well defined since if $I \cong I'$ then $I \otimes_R J \cong I' \otimes_R J$.

0.0.13 Definition

$I \in Mod(R)$ is **invertible** if $\exists J$ with $I \otimes_R J \cong R$ (we call these invertible because $Mod(R)$ under $\otimes$ has unit R).

0.0.14 Theorem

For R a ring of integers, the invertible modules all isomorphic abstractly to some ideal, and so to classify them, it's enough to go over the ideals.

Part IV
Problems

- 30: Show that if R is an integral domain that $I \cong J$ as R-modules if and only if there is α with $\alpha I \cong J$.

- 31: Show that if A is an ideal in R and αA is principal for some $\alpha \in R$, then A is principal. Conclude that principal ideals form an ideal class.
- 32: Show that the ideal classes in R form a group iff for every ideal A there is an ideal B such that AB is principal.

Part V
Bonus- ideals as one dimensional modules.

In the first example, the simplest sheaves are the line bundles on a manifold.

In the second example, the simplest representations are the **characters** of a group, the one dimensional representations.

In the third example, what are the "one dimensional" representations?

0.0.15 Definition

For $M \in Mod(R)$, the **generic rank** of M is the dimension of $M \otimes_R Frac(R)$ as a $Frac(R)$ vector space.

In fact, for every $\mathfrak{p} \subset R$ a prime ideal, the **rank** of M at $\mathfrak{p}$ is the rank of $M \otimes_R Frac(R/\mathfrak{p})$ as a $Frac(R/\mathfrak{p})$ vector space. The **generic rank** corresponds to $\mathfrak{p} = 0$.

0.0.16 Examples

For $R = \mathbb{Z}$, here are some modules with their ranks:

1. $\mathbb{Z}$, has rank 1 at every $\mathfrak{p}$.
2. $\mathbb{Z} \oplus \mathbb{Z}$, has rank 2 at every $\mathfrak{p}$. The rank is additive.
3. $(p\mathbb{Z})$, has rank 1 at every $\mathfrak{p}$, it is isomorphic to $\mathbb{Z}$ abstractly.
4. $\mathbb{Z}/p = \{x/p, x \in \mathbb{Z}\} \subset \mathbb{Q}$ has rank 1 at every $\mathfrak{p}$, and is isomorphic to $\mathbb{Z}$ abstractly.
5. $\mathbb{Q}$, has rank 1 at $\mathfrak{p} = 0$, and rank 0 at $\mathfrak{p} \neq 0$ (since $\mathbb{Q}/p = 0$).
6. $\mathbb{Q}/\mathbb{Z}$, has rank 0 everywhere.
7. $\mathbb{F}_p$ is rank 1 precisely at $\mathfrak{p} = \mathbb{F}_p$.
8. $\mathbb{Q} \oplus \mathbb{F}_2 \oplus \mathbb{F}_3 \oplus \mathbb{F}_5 ...$ is rank 1 at every prime $\mathfrak{p}$.
9. If I, J are ideals, then $rank(I \otimes J)_{\mathfrak{p}} = rank(I)_{\mathfrak{p}} \times rank(J)_{\mathfrak{p}}$. Therefore, every invertible module is of rank 1 everywhere, but example 8 shows that's not an iff.

0.0.17 Footnote

In the unified language of algebra and geometry combining $1, 3$, every ideal of R corresponds to a **line bundle** on $Spec(R)$ (A geometric object attached to R), a module that "locally" is isomorphic to R itself. Conversely, every line bundle is isomorphic to some ideal (At least if $Spec(R)$ is smooth).

0.0.18 Remark

We've seen in example 8 that checking the rank to be 1 isn't enough to detect invertible ideals.

However example 8 is monstrous, by adding a requirement of being finitely generated, it is true that if R is a ring of integers then that implies the module is invertible.

0.0.19 Theorem

For R a ring of integers, the invertible modules all isomorphic abstractly to some ideal, and so to classify them, it's enough to go over the ideals.

0.1 What's so special about R a ring of integers

$\mathcal{O}_K$ will turn out to be a Dedekind domain, which means $Spec(R)$ is a smooth curve.

On a nonsmooth curve with singularity p, $\mathcal{O}(-p)$ is a non-invertible ideal.

Part VI
References

Marcus number fields.